National University

of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

Department of Software Engineering

Course Title	SNTP
Assignment	2
Due Date	Tuesday, 20 October 2026, 10:00 PM

Instructions:

Required Naming Convention

1. ZIP archive: RollNumber_SNTP_Assignment.zip
2. Single outer folder inside the ZIP: RollNumber_SNTP_Assignment
3. Replace RollNumber with your own roll number in both names. For example, 22F3143_SNTP_Assignment.zip must contain the folder 22F3143_SNTP_Assignment.

Required submission contents

1. Runnable source code covering all eight requirements, including data preparation and experiment execution.
2. The dataset or reproducible instructions for obtaining or generating it. If the full dataset is too large to include, provide a small runnable sample and explain how to obtain the full version.
3. A dependency list with versions, saved experiment settings, random seeds, and any configuration needed to reproduce the results.
4. A written report explaining the dataset, assumptions, algorithm choices, implementation, evaluation design, results, computational cost, optimization, limitations, and final conclusion. Submit it as a searchable PDF.

Root README requirements

1. Place a file named README.md directly inside the outer submission folder. It must provide:
2. Your name and roll number, and a map showing where the source code, dataset, configuration, results, and report are located.
3. The required software environment, dependency installation instructions, and exact commands for preparing data and running the solution from the outer folder.
4. Instructions for running a small demonstration and reproducing the full experiments, with expected runtime and hardware requirements.
5. A mapping from each of the eight assignment requirements to the relevant source files or functions and report sections, plus any known limitations.

Recommendation & Information Diffusion in a Social Network

Scenario

A social platform wants to recommend useful content to its users while understanding how those recommendations influence the spread of information. Users have different interests, interact with different people, and may adopt content because they find it relevant or because their friends have adopted it.

A recommendation that attracts many users may not satisfy their individual preferences. Similarly, a highly connected user may not be the best starting point for spreading content if their connections overlap heavily with those of other selected users.

How would you design, implement, and experimentally evaluate a system that jointly considers personalized recommendation, social influence, and information diffusion?

Develop a working solution addressing all eight requirements below. Select and justify your methods using the concepts covered in the lecture slides. Support your decisions with reproducible experiments and explain what the results mean.

1. Represent the social environment

Use a suitable public dataset or construct a reproducible synthetic dataset containing users, social connections, items, and user–item interactions. Support different user preferences, incomplete interaction histories, and varying strengths of social influence.

Explain the meaning and direction of connections, how interactions become preference signals, and any assumptions introduced when combining or generating data. Distinguish observed information from information that you create or infer.

2. Learn preferences and generate recommendations

Implement two substantially different recommendation approaches from the studied material. At least one must learn from user–item interactions. Both approaches must predict preferences for unseen items and generate ranked recommendations.

Investigate their behavior when users have limited histories, items receive few interactions, or rating patterns differ across users. Implement the central recommendation logic yourself. Libraries may support numerical operations and data handling but must not replace your recommender with a ready-made implementation.

3. Incorporate social relationships

Extend one recommendation approach to use the social network. Investigate whether friends provide more useful preference information than users who are similar but socially unconnected. Allow the contribution of social information to change so its effect can be isolated experimentally. Explain when social information improves recommendations and when it makes them worse.

4. Model adoption and information spread

Allow recommended items to initiate diffusion through the network. Distinguish receiving a recommendation, becoming exposed through a neighbor, and actually adopting an item.

Implement the Independent Cascade Model and one epidemic model studied in the slides. Explain what the states and transitions mean in your platform and why the models may produce different outcomes. Run repeated simulations to investigate how preferences, network structure, and influence strength affect the size and speed of diffusion.

5. Select users under a limited promotion budget

Assume the platform can initially promote an item to only a limited number of users. Develop a strategy for selecting this initial group and compare it against random selection and a simple network-based baseline under the same budget.

Your strategy must consider the additional spread contributed by a candidate rather than relying only on individual popularity. Investigate how overlapping audiences and changes in the promotion budget affect results. Account for uncertainty when estimating a candidate's contribution.

6. Recommend to groups

Extend the system to recommend an item to a group whose members have conflicting preferences. Compare average satisfaction, least misery, and most pleasure. Analyze whether the item preferred by the group-recommendation strategy is also the item that spreads most widely when introduced through that group.

7. Evaluate the complete system

Design experiments that separately assess recommendation quality, diffusion behavior, and their interaction. Use appropriate measures of prediction error, precision, recall, ranking quality, coverage, diversity, and diffusion outcomes. Include accuracy only where you define a meaningful classification task and explain what counts as a negative example.

Prevent training–test leakage. Explain your data split, relevance definition, recommendation cutoff, and candidate-item selection. Keep observed outcomes separate from outcomes generated by your simulator, and explain what conclusions each can support. Use repeated trials and report variability rather than relying on a single simulation.

8. Investigate intervention and computational cost

Introduce an intervention that limits diffusion and examine its effects on both spread and user satisfaction.

Evaluate how runtime and memory requirements change as the network, item collection, and promotion budget grow. Identify the main computational bottleneck and implement an optimization whose effect you measure.

Final discussion

Under what conditions does combining recommendation with social influence improve the platform's outcomes, and when does it create a trade-off between individual satisfaction and widespread adoption?

Answer this question using evidence from your experiments. Discuss limitations, unexpected results, and cases where your approach performs poorly.